



Research report

Haloperidol blocks dorsal striatum activity but not analgesia in a placebo paradigm



Nathalie Wrobel^{a,c,*}, Katja Wiech^b, Katarina Forkmann^c,
Christoph Ritter^{a,c} and Ulrike Bingel^c

^a Department of Neurology, University Medical Center Hamburg-Eppendorf, Hamburg, Germany

^b Oxford Centre for Functional Magnetic Resonance Imaging of the Brain, Nuffield Department of Clinical Neurosciences, John Radcliffe Hospital, University of Oxford, Oxford, UK

^c Department of Neurology, University Duisburg-Essen, Germany

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ABSTRACT

Although placebo analgesia has been associated with an engagement of the endogenous opioid system there is growing evidence from neuropharmacological studies for an involvement of additional neurotransmitter systems. An increased dopaminergic neurotransmission in the ventral basal ganglia that has been found during placebo analgesia suggests a role for the dopaminergic system (Scott et al., 2007). It is, however, unclear whether striatal dopaminergic activity is causally involved in this type of analgesia. This study aimed at exploring the functional role of the dopaminergic system in placebo analgesia. To this end, we investigated the effect of the dopamine D2/D3 receptor antagonist haloperidol on behavioral and neural measures of placebo analgesia using functional magnetic resonance imaging (fMRI) in healthy volunteers. We found that 2 mg haloperidol p.o. significantly reduced the correlation between dorsal striatum activity and the individual placebo response, but had no significant effect on placebo analgesia at the behavioral or neural level, as indexed by activity in sensory or pain-modulatory brain regions.

Our study therefore suggests that dopaminergic neurotransmission might not be causally involved in placebo analgesia but is related to phenomena associated with placebo analgesia such as reward processing and learning.

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1. Introduction

Placebo analgesia represents a prime example of endogenous pain modulation and impressively demonstrates our ability to inhibit ascending nociceptive information under certain

conditions. Evidence from neuroimaging studies indicates that placebo analgesia involves a top-down activation of endogenous analgesic mechanisms via the descending pain-modulatory system at all levels of the central nervous system, including the spinal cord (Bingel, Lorenz, Schoell, Weiller, & Buchel, 2006; Eippert et al., 2009; Petrovic, 2005; Wager et al.,

* Corresponding author. Department of Neurology, University Medical Center Hamburg-Eppendorf, Martinistr. 52, D-20246 Hamburg, Germany.

E-mail address: n.wrobel@uke.de (N. Wrobel).

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Table 1 – Participants demographics and questionnaire results.

	Haloperidol	Saline		
N	21	17	38	
Age	24.2 ± 1.3	23.5 ± 1.4	23.7 ± 1.0	
Gender (women)	9	7	16	
Center for Epidemiological Studies Depression Scale	11.6 ± 0.59	12.07 ± 1.15	t(36) = −.386	n.s.
State–Trait Anxiety Inventory				
State	43.8 ± 1.2	45.95 ± 2.43	t(36) = −.828	n.s.
Trait	37.4 ± 1.86	35.2 ± 2.5	t(36) = .721	n.s.
Crowne–Marlowe Social Desirability Scale	11.85 ± 0.87	11.87 ± 1.17	t(36) = −.012	n.s.

2004). Activity in this pathway is complemented by intra-cortical mechanisms, predominantly involving limbic and paralimbic regions (Wager, Atlas, Leotti, & Rilling, 2011).

Both Positron emission tomography (PET) studies using μ -opioid sensitive tracers (Scott et al., 2008; Wager, Scott, & Zubieta, 2007; Zubieta et al., 2005) and pharmacological studies using the opioid antagonist naloxone (Amanzio & Benedetti, 1999; Eippert et al., 2009; Grevert, Albert, & Goldstein, 1983; Levine & Gordon, 1984; Levine, Gordon, & Fields, 1978) have shown that the endogenous opioid system is an important mediator of placebo analgesia. However, neuropharmacological studies also suggest an involvement of additional, non-opioidergic components of placebo analgesia (Amanzio & Benedetti, 1999). Functional molecular imaging investigating changes in the binding potential of carbon 11 [^{11}C]-labeled raclopride has shown an increased dopaminergic neurotransmission in the nucleus accumbens (NAC), putamen and caudate that correlated with the individual placebo-analgesic response (Scott et al., 2007). Furthermore, Schweinhardt et al. reported a close positive relationship between gray matter density in the ventral striatum and the magnitude of placebo analgesia as well as dopamine-related personality traits using voxel-based morphometry (Schweinhardt, Seminowicz, Jaeger, Duncan, & Bushnell, 2009). These findings suggest a potentially relevant role of the dopaminergic system and specifically the striatum in placebo analgesia. It is, however, unclear whether striatal dopaminergic activity is causally involved in the generation of this type of analgesia.

To explore the role of the dopaminergic system in placebo analgesia, we performed pharmacological functional magnetic resonance imaging (fMRI) (Honey & Bullmore, 2004; Leslie & James, 2000; Schweinhardt et al., 2009) using a classical placebo heat pain paradigm in healthy volunteers who were randomly assigned to either receive the D2/D3-antagonist haloperidol or an inert substance in a double-blind design. We used an established placebo analgesia paradigm that comprised an initial expectation manipulation (conditioning) phase and a later test phase.

2. Material and methods

2.1. Subjects

50 healthy volunteers (mean age: 26.56 years; range: 22–34; 23 females) participated in this study. All subjects had heat pain

thresholds in a normal range at the site of stimulation and had no known history of neurological or psychiatric diseases, including recurrent or chronic pain. The study was approved by the local Ethics committee of the Medical Board in Hamburg, Germany and all experimental procedures conformed to the Declaration of Helsinki. All participants gave written informed consent to the experimental procedures and were free to withdraw from the study at any time. Please note that the participants were partially deceived to the nature of the study (i.e., placebo manipulation).

Participants were randomly assigned to one of two groups. The experimental group received the dopamine antagonist haloperidol (*haloperidol group*) while the control group received saline (*saline group*), see Section 2.4 for details.

Twelve participants did not complete the study or were excluded after scanning due to the following reasons: technical problems with the thermal stimulation ($N = 3$), lack of compliance during or failure of temperature calibration ($N = 3$), excessive movement during the scanning session [$N = 2$; >10 mm initial misalignment (summed across x/y/z dimensions) (Ardekani, Bachman, & Helpner, 2001)] and incidental findings in their anatomical scans (large cysts $N = 4$ that impaired an adequate preprocessing of the images). Data from the remaining 38 participants (21 in the haloperidol group, 17 in the saline group) were included in the final behavioral and neuroimaging analyses.

The groups did not differ significantly with regard to age, gender distribution, weight (mean weight 75 kg), psychological scores (depression, anxiety, social desirability), or pain thresholds. For participants' characteristics see Table 1.

2.2. Experimental paradigm

We used a 2×2 mixed-factorial design with the between-subject factor group (*haloperidol vs saline*) and the within-subject factor condition (*control vs placebo*) in a well-established placebo heat pain paradigm involving both expectation and conditioning components (Colloca & Benedetti, 2006; Eippert et al., 2009; Montgomery & Kirsch, 1997; Voudouris, Peck, & Coleman, 1990; Wager et al., 2004). In this paradigm the expectation of pain relief is induced by applying a supposedly analgesic cream which is de facto inert. The expectation manipulation (conditioning) procedure mimics the analgesic effect of the analgesic cream by manipulating the temperature applied to the placebo site and has been shown to strengthen the expectation of pain relief.

The experiment took place on two days with one day between sessions. Day 1 included an introductory session, determination of pain thresholds, calibration of stimulus intensities, the first conditioning session and filling in of questionnaires. Day 2 comprised determination of pain thresholds, the second conditioning session, oral application of haloperidol or saline and 3.5 h of rest which was followed by the actual test phase performed in the MR scanner (fMRI test session) and debriefing.

On day 1, experimenter (N.W.) welcomed each participant in a white lab coat, informed about the experimental procedures and obtained written informed consent for the study. All instructions were read to the participants to ensure a standardized procedure.

For the stimulation, five 4×4 cm squares were drawn on the participants' left volar forearm. The two upper or lower squares were outlined in green or red and designated as the site for later placebo or control cream stimulation, respectively. As in our previous study (Eippert et al., 2009) we used two patches for each condition to prevent sensitization through stimulation of the same patch during the conditioning and test phase. The coloring (green = placebo, red = control) was chosen to reinforce the association between the skin patch and pain relief. The assignment of the placebo or control cream to the upper or lower patches was randomized across participants, such that half of the participants received the placebo cream on the upper patches and the other half received placebo cream on the lower patches. Likewise, the order of stimulation (i.e., placebo or control site) in the conditioning session was randomized across subjects. The fifth square located between the upper and lower squares served as a neutral site where heat pain thresholds were assessed prior to each of the two sessions using the method of limits (Fruhstorfer, Lindblom, & Schmidt, 1976). Subsequently, participants were familiarized with the equipment and rating procedure by presenting stimuli of different temperature levels to the left forearm that were rated through button presses of an optical USB mouse on a visual analog scale visible on the computer screen [VAS (100 parts; endpoints labeled with 'no pain' and 'unbearable pain')].

We then applied two identical looking, inert creams which were introduced as a local anesthetic (placebo) and a control cream (control) and were kept in professionally labeled tubes. The tube of the placebo cream and control cream were marked in green or red respectively to reinforce the association with the placebo or control site. Both creams were applied to one of the two green or red squares respectively for 15 min during which a calibration procedure was performed on the middle (neutral) square.

The calibration procedure was performed to determine the individual temperatures to evoke pain levels of VAS 40, 60, and 80. To this end, a pseudo-random sequence of 16 thermal stimuli (duration 20 sec) with different intensities around the participants' individual pain threshold (range: $-1^\circ\text{C} < \text{pain threshold} < +3.5^\circ\text{C}$, temperature difference of $.5^\circ\text{C}$, each temperature was presented twice) was applied. Subsequently, the participants were asked to rate the intensity of each stimulus on a VAS. The individual temperatures resulting in VAS ratings of 40, 60, and 80 were determined using a linear

regression of the calibration ratings. This procedure ensured a perception-locked stimulation across subjects.

2.2.1. Conditioning day 1

After the two creams had been removed, participants underwent a conditioning procedure to reinforce the credibility of the placebo manipulation. Twelve heat stimuli of 20 sec duration were applied to both the placebo and control site and subjects rated the pain intensity on the VAS following each stimulus. Each trial consisted of five phases: anticipation, pain, pause, rating and rest. The anticipation phase commenced when the color of the crosshair that was presented on the computer screen changed from white to red, indicating that a painful stimulation would follow shortly. Subjects had to press a button as quickly as possible when the crosshair changed color. This procedure ensured that alertness was maintained by all participants throughout the experiment. After a variable delay (7.5 ± 3.5 sec), a 20 sec painful thermal stimulus was administered (baseline temperature: 35°C ; ~ 1.5 sec ramp up, 17 sec plateau, ~ 1.5 sec ramp down). A variable delay (5 ± 2 sec) followed the thermal stimulation before subjects had to rate the level of pain during the trial using a VAS presented on the computer screen. A variable inter-trial interval (ITI; 20 ± 5 sec) followed during which a white crosshair was displayed (Fig. 1).

During the conditioning session participants expected to receive pain stimuli individually calibrated at a pain level of VAS 80 on both skin patches (placebo and control site). Unbeknownst to them the temperature was lowered to correspond to a pain level of VAS 40 for stimuli applied to the placebo site. This procedure served to enhance the participants' expectation that the placebo cream was an effective analgesic. The stimulation intensity on the control site was kept at a level corresponding to a VAS of 80. Prior to each session pain thresholds were reassessed using the method of limits. In order to further strengthen the impression of an analgesic effect, the rise time of the thermode on the placebo site was decreased from $1.0^\circ\text{C}/\text{sec}$ on the control site to $.7^\circ\text{C}/\text{sec}$ on the placebo site.

2.2.2. Questionnaires

At the end of the first day subjects filled in an electronic version of the following questionnaires: Center for Epidemiological Studies Depression Scale (Hautzinger, 1993), State-Trait Anxiety Inventory (Spielberger, 1970), and Crowne-Marlowe Social Desirability Scale (Crowne & Marlowe, 1960).

For day 2, participants were instructed to refrain from eating 2 h prior to the experiment. Upon arrival, subjects were first tested for recent drug use (including THC, opiates and amphetamines) using a commercially available urine test (Diagnostik-Nord, Schwerin, Germany).

Subsequently, the placebo and control creams were applied on the respective skin areas, which were marked according to the previous testing. During a waiting period of 15 min in which the 'analgesic' cream was expected to unfold its effect, the heat pain threshold was assessed on the neutral site. Further, six test stimuli with temperatures corresponding to VAS 40, 60 and 80 as determined on day 1 were applied to test for possible variations in pain sensitivity across the

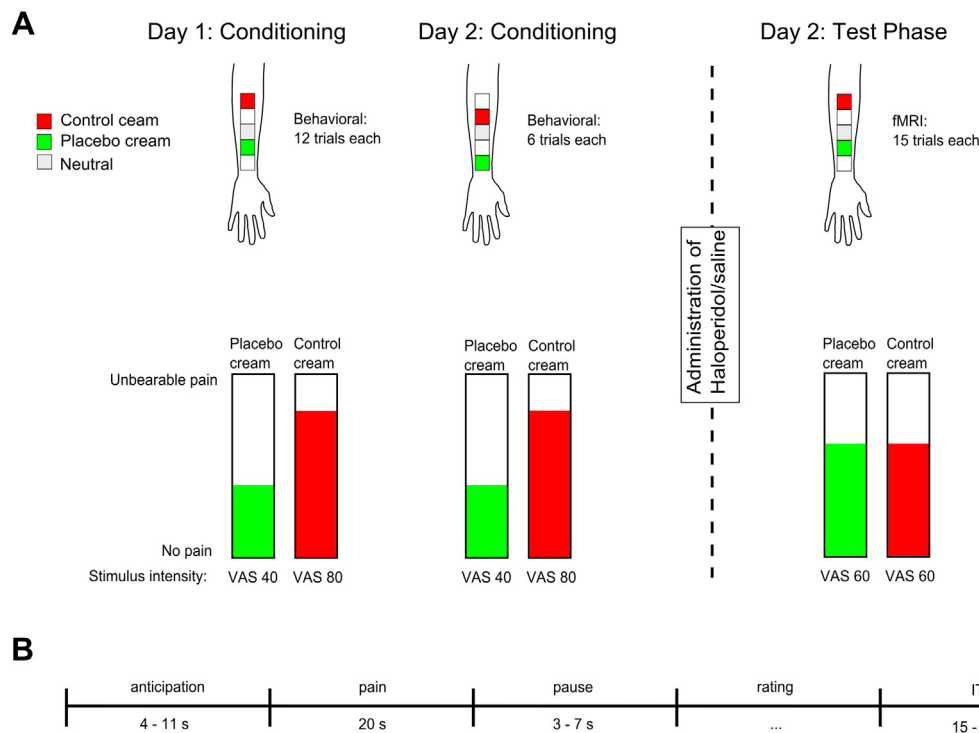


Fig. 1 – Experimental design: (A) The experiment took place on two consecutive days and consisted of three phases: conditioning on day 1, conditioning on day 2, and test on day 2. At the beginning of each phase, subjects were treated with two identical creams on their left forearm. Subjects were instructed that one cream was a highly effective analgesic whereas the other served as a sensory control. During the conditioning phases (which consisted of 12 trials on the placebo-treated site and 12 trials on the control site in conditioning phase 1 and 6 trials each in conditioning phase 2) the painful stimulation on the placebo-treated site was surreptitiously lowered from VAS 80 to VAS 40 to reinforce the subjects' expectation that they had received a potent analgesic cream. 3.5 h before the test phase, subjects received either an oral application of 2 mg haloperidol or saline as a tablet to account for pharmacokinetics of haloperidol. fMRI data were collected during the test phase which consisted of 15 trials in each condition. Importantly, the intensity of the painful stimulation was identical on both skin patches (VAS 60) in order to test for placebo-analgesic effects. **(B)** Each trial consisted of five phases: anticipation phase, pain, a short pause, rating, and an ITI. The anticipation phase began when the white crosshair that was displayed on the computer screen turned into a red crosshair, indicating that a painful stimulation would follow shortly. Subjects had to press a button as quickly as possible when the crosshair changed color. After a variable delay the painful thermal stimulus (duration 20 sec) was administered, 3–7 sec after the thermal stimulation subjects had to rate the pain intensity using a VAS. The trial was completed by a 15–25 sec baseline during which a white crosshair was displayed.

experimental sessions. If the pain sensations evoked by the temperatures from day 1 differed by more than 10% a recalibration was performed.

2.2.3. Conditioning day 2

The subsequent second conditioning phase involved the same procedures as outlined above, except that only six stimuli were applied to each site (placebo and control). This second conditioning session served to reactivate and strengthen expectations of pain relief due to the placebo cream (Colloca & Benedetti, 2006; Colloca, Sigauco, & Benedetti, 2008).

Upon completion of the second conditioning phase 2 mg encapsulated haloperidol or saline were administered by experimenter T.F. using a double-blind procedure (also see Section 2.4 below). This was done to ensure that the initial expectation and conditioning related mechanisms were not

confounded by the manipulation of dopaminergic transmission. Subjects were then accompanied to a waiting room where light snacks and drinks were offered after an additional 60 min. After 3.5 h, subjects were led to the fMRI facilities where they completed two questionnaires assessing potential adverse effects of the test medication (haloperidol/saline) and their current mood before the actual test session started.

2.3. Test phase

Commencing the fMRI test session placebo and control creams were reapplied to the respective skin areas, which had been color-coded (green = placebo, red = control) as during the conditioning phase. After a waiting period of 15 min to allow 'the local anesthetic to become effective', subjects were positioned in the scanner. The test phase inside the scanner

consisted of a placebo and a control session (with 15 trials each) which were presented in a randomized order across subjects. Importantly, placebo and control sites were stimulated with the same individually calibrated temperature corresponding to VAS 60. The placebo effect was defined as the difference in perceived pain intensity on the placebo-treated compared to the control site during the test phase where identical stimulation intensities are applied to both sites.

2.4. Drug administration

Subjects were randomly assigned to either receive 2 mg haloperidol or saline (as capsules) using a double-blind, 2×2 design with the within-subject factor **condition** (placebo vs control) and the between-subject factor **group** (haloperidol vs saline). The dosage and application of haloperidol were guided by previous studies demonstrating a successful modification of reward-related responses (behaviorally and neurally) in healthy volunteers using similar dosing regimens. These include a sensory decision-making paradigm using 2 mg of haloperidol (Pleger et al., 2009), an instrumental reinforcement task involving monetary gains and losses using 1 mg of haloperidol (Pessiglione, Seymour, Flandin, Dolan, & Frith, 2006), and the subconscious processing of sexual stimuli using 3 mg of haloperidol (Oei, Rombouts, Soeter, van Gerven, & Both, 2012). Seventeen subjects provided blood samples on a voluntary basis approximately 3.5 h after haloperidol or placebo intake to determine haloperidol plasma levels.

Participants were informed about the pharmacological properties, general clinical use and potential side effects of haloperidol. Furthermore, participants were informed that the purpose of the study was to explore dopaminergic mechanisms of pain perception and pain modulation. The two experimenters who interacted with the subjects (T.F. administered the drug; N.W. carried out the experiments) were blinded as to which drug was given. U.B. who did not interact with the subjects was responsible for the assignment of the participants to the two groups. The group assignment was only unblinded after the last participant had completed the experiment.

The assessment of our pharmacological manipulation with 2 mg haloperidol, mood and physical side effects were assessed 3.5 h after haloperidol or saline administration.

Subjects filled in a 7-item questionnaire regarding physical side effects which has been used in previous pharmacological imaging studies (Bentley, Husain, & Dolan, 2004; Pessiglione et al., 2006; Petrovic, 2005). Each adverse effect could be rated as 'inexistent', 'very weak', 'weak', 'moderate', 'strong', 'very strong', and 'extremely strong' (scores: 0–6). Subjects also rated their current mood on a 16-item questionnaire which has also been employed in previous pharmacological imaging studies (Bentley et al., 2004; Pessiglione et al., 2006; Petrovic, 2005). Each item consisted of a VAS with opposing verbal descriptors at each end. Ratings were scored from 1 to 100. Following the procedure described by Bond and Lader (1974) the 16 scales were log-transformed and skewed scores were reversed. Next, scales were grouped into three categories: 'alertness' (ranging from 'alert' to 'drowsy'), 'calmness' (ranging from 'relaxed' to 'tensed') and 'contentment' (ranking from 'contented' to 'discontented').

2.5. Data acquisition

Behavioral data (VAS pain ratings, reaction times) were automatically recorded and logged by the stimulation program Presentation® (Neurobehavioral Systems, Albany, CA, USA) used for stimulus presentation. Thermal stimulation was carried out using a thermode (40×40 mm Peltier device; TSAII, Medoc, Tel Aviv, Israel).

fMRI data were acquired on a 3 T system (Siemens Magnetom Trio, Siemens, Erlangen, Germany) equipped with a 12 channel head coil. Forty-four transversal slices (slice thickness: 2 mm, gap: 1 mm, slice order: descending) were acquired in each volume (repetition time: 2.6 sec, echo time: 25 msec, flip angle: 90° , field of view: 208×208 mm, matrix: 104×104 ; GRAPPA with PAT-factor 2 and 48 reference lines) using a T2*-weighted gradient-echo echo-planar imaging sequence (EPI). Slice orientation was tilted by -39° , which allowed for coverage of areas as ventral as the medulla. The first six volumes of each session were discarded to allow for T1 saturation. We also acquired high-resolution ($1 \times 1 \times 1$ mm voxel size) T1-weighted images for each subject using an MPRAGE sequence.

2.6. Data analysis

All behavioral data were analyzed with SPSS 13. Pain ratings were analyzed using a mixed two-way analysis of variances (ANOVAs), with the between-subject factor group (haloperidol vs saline) and the within-subject factor condition (control vs placebo). We used separate ANOVAs to analyze pain ratings during the manipulation and test sessions. To test for differences between treatment groups in pain thresholds on the control site, we performed an ANOVA with the between-subject factor group (haloperidol vs saline) and within-subject factor time (pre- and post-haloperidol or saline application).

A two-sample t-test was performed to test for group differences in the three mood factors and the sum score of the adverse effects questionnaire. We also compared the temperatures in the test phase on the control and placebo sites to reach VAS 60 between groups, to assure physically identical stimulation across groups. In all tests, results surviving a threshold of $p < .05$ were considered as statistically significant. Unless indicated otherwise results are presented in mean $\pm$ standard error of the mean.

2.7. Image processing and statistical analysis

Image processing and statistical analysis of fMRI data were performed using SPM8 [<http://www.fil.ion.ucl.ac.uk/spm/>] (Friston et al., 1995; Worsley & Friston, 1995)]. After removing the first 6 volumes to compensate for T1 saturation effects, preprocessing included slice timing (correction for differences in slice acquisition time), realignment to the first volume (motion correction), unwarping (correction for interaction of movements and field inhomogeneities), normalization to standard Montreal Neurological Institute (MNI) space and finally smoothing with an 8 mm Gaussian kernel with full-width at half-maximum (FWHM). Data were also subjected to high-pass filtering (cut-off period: 128 sec) and correction for temporal autocorrelations (based on a first-order autoregressive model).

Data analysis was performed using a general linear model approach. For the fMRI analysis the 20 sec painful stimulation period was divided into an early and a late phase based on previous studies (Eippert et al., 2009; Wager et al., 2004).

Accordingly, the first-level design matrix included ten regressors (five in each session): *anticipation placebo*, *anticipation control*, *pain-early placebo*, *pain-early control*, *pain-late placebo*, *pain-late control*, *rating placebo*, *rating control*, and two session constants. Anticipation was modeled by convolving a delta function (at anticipation onset) with the canonical hemodynamic response function (HRF). *Pain-early* was modeled by convolving a 10 sec boxcar function (starting at pain onset) with the canonical HRF. *Pain-late* was modeled by convolving a 10 sec boxcar function (starting 11 sec after pain onset) with the canonical HRF, and rating was modeled by convolving delta functions (representing each button press during rating) with the canonical HRF.

Following model estimation, the resulting first-level contrast images from each subject were used for the second-level analysis. Second-level design matrices were configured using SPM's 'flexible factorial' model and included four regressors (i.e., saline group: pain-early placebo, saline group: pain-early control, haloperidol group: pain-early placebo, haloperidol group: pain-early control). A random effects approach (Friston, Holmes, Price, Buchel, & Worsley, 1999) was used and we corrected for possible non-sphericity of the error term [unequal subject variance, dependence of conditions and possible unequal variance between groups (Gläscher & Gitelman, 2008)].

In a first analysis, we tested for regionally specific neural placebo effects and their modulation by haloperidol (main effect of condition and interaction). We thus used the main effects [*control* > *placebo*] and [*placebo* > *control*] to test for placebo-associated changes in neural activity and the interaction terms [*saline*_[control>placebo] > *haloperidol*_[control>placebo]] and [*saline*_[placebo>control] > *haloperidol*_[placebo>control]] to test for group-by-condition interactions. These analyses were carried out separately for early pain and late pain. Due to a lack of *a priori* hypothesis regarding the temporal dynamics of the effect of haloperidol on placebo-related group-by-interaction activity, the entire pain period was used for the interaction analysis. As in previous studies (Scott et al., 2007; Wager et al., 2007), this analysis was restricted to placebo responders ($N = 29$, haloperidol group: $N = 15$) to increase sensitivity for dopamine-dependent changes in placebo-related activity. Responders were defined those participants who exhibited a pain reduction of at least 10 points on the VAS on the placebo compared to the control site [*control rating*–*placebo rating* ≥ 10].

Furthermore, we tested for global effects of haloperidol on pain-related neural activity (main effect of group) across the entire period of noxious stimulation using the contrasts [*haloperidol* > *saline*] and [*saline* > *haloperidol*]. As SPM's 'flexible factorial model' does not calculate group and subject variance separately in a mixed model, the main effect of group was not derived from the ANOVA but from a two-sample t-test on the first-level contrasts [*pain-early control*, *pain-late control*]. This analysis was performed on all $N = 38$ participants included in the final data analyses. For congruency with the other analyses Table 4 additionally reports the main effect of group for the $N = 29$ placebo responders.

We report p -values at a level of $p < .05$, either correcting for the entire brain (familywise error (FWE) correction) or for *a priori* selected regions of interest (Small Volume Correction – SVC). These include brain areas known to be involved in pain processing (thalamus, posterior insula, somatosensory and mid-cingulate cortex) and its modulation (periaqueductal gray – PAG, prefrontal cortex, rostral anterior cingulate cortex – rACC, as well as the striatum including the NAc). Regions of interest were defined as spheres centred around the peak coordinates (ignoring laterality) obtained from previous studies. Cortical areas were corrected using a sphere of 20 mm and included the following areas: the dorsolateral prefrontal cortex (DLPFC) (Faymonville et al., 2000), the rACC (Zubieta et al., 2005) and the orbitofrontal cortex (OFC) (Wager et al., 2004), the primary somatosensory cortex (Bingel, Schoell, Herken, Buchel, & May, 2007), the secondary somatosensory cortex (Xu et al., 1997), the posterior insula (Bingel et al., 2007) and the dACC (Frankenstein, Richter, McIntyre, & Remy, 2001). Subcortical areas were corrected using a sphere of 15 mm and involved the striatum [caudate nucleus (Marsh et al., 2010), putamen (Bingel, Glascher, Weiller, & Buchel, 2004), NAc (Scott et al., 2008)] and the thalamus (Wager et al., 2004). For exploratory reasons we also report results from a more lenient threshold of $p \leq .001$ (uncorrected).

3. Results

3.1. Behavioral results

3.1.1. Conditioning procedure

To confirm that our pain intensity manipulation was successful and that groups did not differ in basic pain sensitivity prior to the pharmacological modulation, ANOVAs were computed on the pain ratings from the manipulation sessions

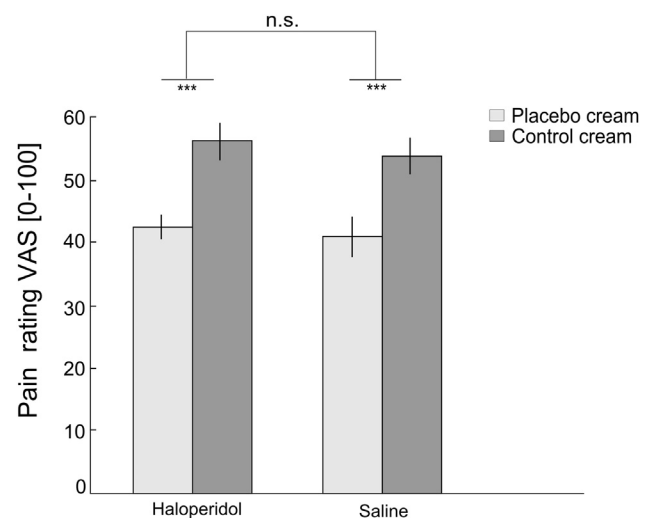


Fig. 2 – Behavioral effects: Pain ratings (mean $\pm$ sem) indicate a significant placebo effect [Δ control–placebo] in the haloperidol and saline group with no difference between groups, suggesting that a blockage of D2/D3 dopamine receptors had no significant effect on the placebo-analgesic response.

Table 2 – Placebo-related BOLD responses in pain-modulatory and pain-responsive cortical and subcortical areas.

Region	Coordinate of peak voxel (x, y, z in mm)		Voxel-level (T)	Number of voxels in cluster
	R: right	L: left	R/L	
(A) Early pain: [placebo > control]				
DLPFC	50, 30, 36		4.5	50
DLPFC	34, 22, 48		3.7	40
rACC	14, 46, 22		3.8	20
OFC	28, 56, −2		3.5	7
(B) Early pain: [control > placebo]				
Putamen	28, 4, 6		3.9	60
Posterior insula/SII	56, −24, 22		3.4	98
Head of caudate		−10, 10, 0	2.9	4
(C) Late pain: [placebo > control]				
DLPFC	38, 20, 34		3.7	10*
OFC	28, 44, −8		3.3	10*
(D) Late pain: [control > placebo]				
SII	44, −42, 24		4.4	21
dACC	16, −6, 38		4.2	8
Posterior insula/SII	28, −24, 18		3.9	13
SI	34, −34, 66		3.8	16
Thalamus		−4, −12, 12	3.5	2
(E) Entire pain period: interaction contrast				
[saline] _[placebo>control] > [haloperidol] _[placebo>control]				
Caudate body		−8, 14, 16	3.9	7
(F) Entire pain period: interaction contrast				
[saline] _[control>placebo] > [haloperidol] _[control>placebo]				
—	—	—	—	—

Changes in BOLD responses to painful thermal stimulation for the indicated contrasts. Coordinates are denoted by x, y, z in mm (MNI-space) and strength of activation is expressed in t-scores [(A)/(B)/(C)/(D) df = 54 and (E)/(F) df = 27] $p < .05$ corrected, using SVC as indicated in Section 2; * $p < .07$ for DLPFC and OFC in (C).

dACC: dorsal anterior cingulate cortex, DLPFC: dorsolateral prefrontal cortex, OFC: orbitofrontal cortex, rACC: rostral anterior cingulate cortex, SI: primary somatosensory cortex, SII: secondary somatosensory cortex, SVC: small volume correction.

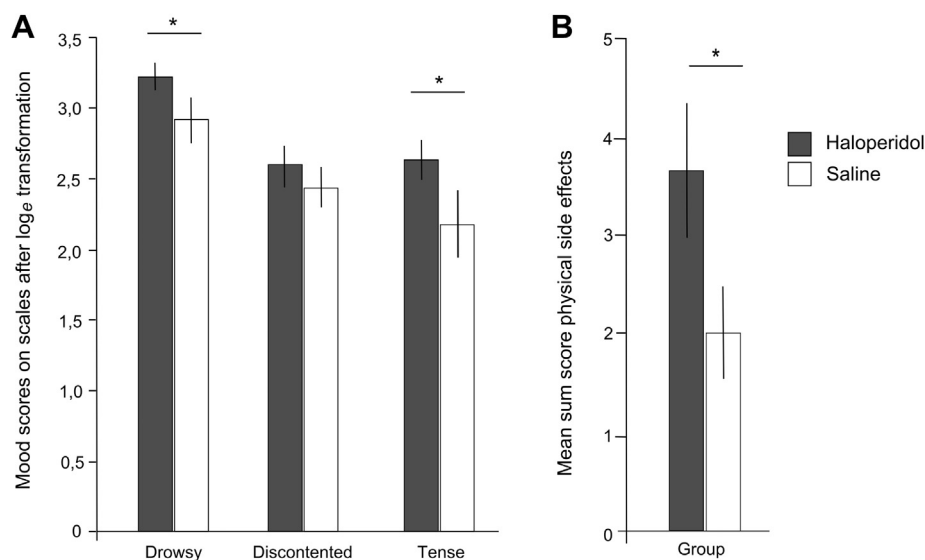


Fig. 3 – Adverse effects of haloperidol: (A) Mood ratings revealed significant mood changes in the haloperidol group compared to the saline group in factors ‘alertness’ (drowsy) and ‘calmness’ (tense). Mood ratings are log-transformed and presented as mean ± sem. (B) The haloperidol group scored significantly higher on the 7-item questionnaire regarding physical side effects (sum score). Bars show mean ± sem.

on day 1 and day 2. Pain ratings reveal that placebo and control sites were rated significantly differently thus indicating a successful manipulation of the pain experience (main effect of condition; day 1: $F_{(1,36)} = 228.98$, $p < .001$; day 2: $F_{(1,36)} = 228.89$, $p < .001$). However, neither a significant main effect of group (day 1: $F_{(1,36)} = .23$, n.s.; day 2: $F_{(1,36)} = .17$, n.s.) nor a significant group-by-condition interaction (day 1: $F_{(1,36)} = .35$, n.s.; day 2: $F_{(1,36)} = .47$, n.s.) was observed, indicating no systematic

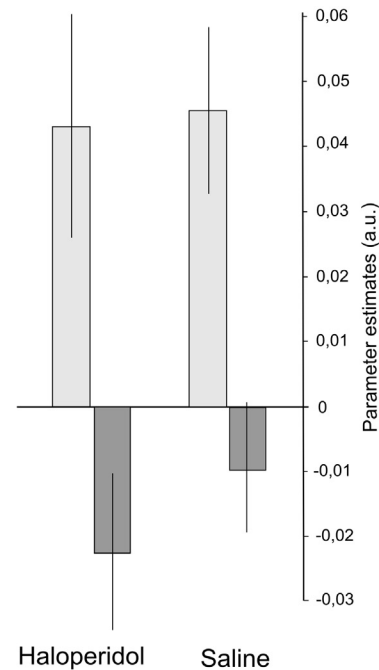
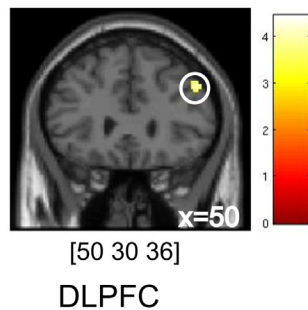
difference in general pain sensitivity or the strength of the conditioning procedure between haloperidol and saline groups.

3.1.2. Pain thresholds and pain sensitivity during test phase

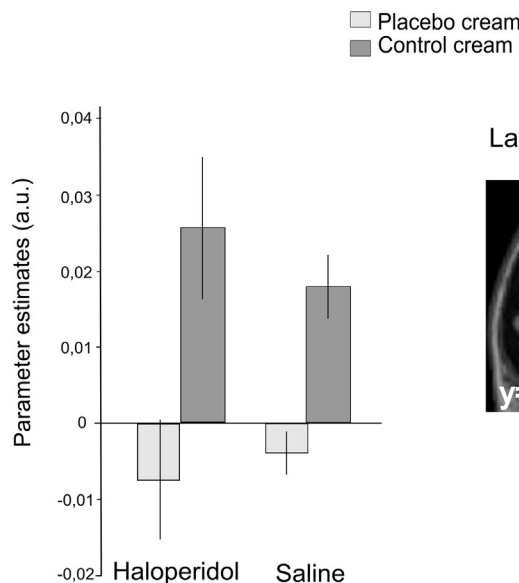
Furthermore, pain thresholds obtained at the beginning of the test phase (heat pain thresholds: $M_{(\text{saline})} = 48.1^\circ\text{C} \pm .4$; $M_{(\text{haloperidol})} = 48.1^\circ\text{C} \pm .5$; interaction: $F_{(1,35)} = .19$, n.s.; group:

A [placebo > control]

Early phase



B [control > placebo]



Late phase

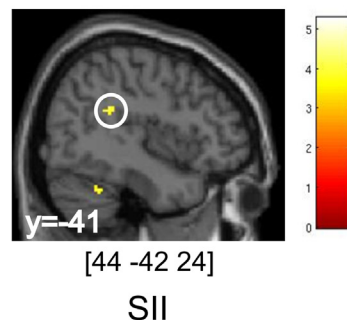


Fig. 4 – Placebo-related BOLD responses in pain-modulatory and pain-responsive cortical and subcortical regions for each group and condition: The analysis is based on placebo responders only ($N = 29$) and shows activation maps of (A) contrast [placebo > control] in the early phase and (B) contrast [control > placebo] in the late phase as well as peak voxel parameter estimates for each group separately. For visualization purposes the image is thresholded at $p < .001$ uncorrected. Color bar indicates t-scores. Error bars indicate standard error of the mean.

$F_{(1,35)} = .4$, n.s.) and temperatures necessary to induce a VAS of 60 on the control and placebo sites (saline 46.8 °C vs haloperidol 46.6 °C; $t_{(33)} = -.8$, n.s.) showed no differences between haloperidol and saline groups. This indicates no significant effect of 2 mg haloperidol on general pain sensitivity. Note that the discrepancy between pain threshold temperatures and the temperatures for a pain level

corresponding to a VAS of 60 is explained by the much longer stimulus duration of the actual test stimuli (20 sec).

3.1.3. Placebo effect

Despite identical thermal stimulation, pain ratings on the placebo site were significantly lower compared to the control site, indicating a significant placebo-analgesic effect (main effect of

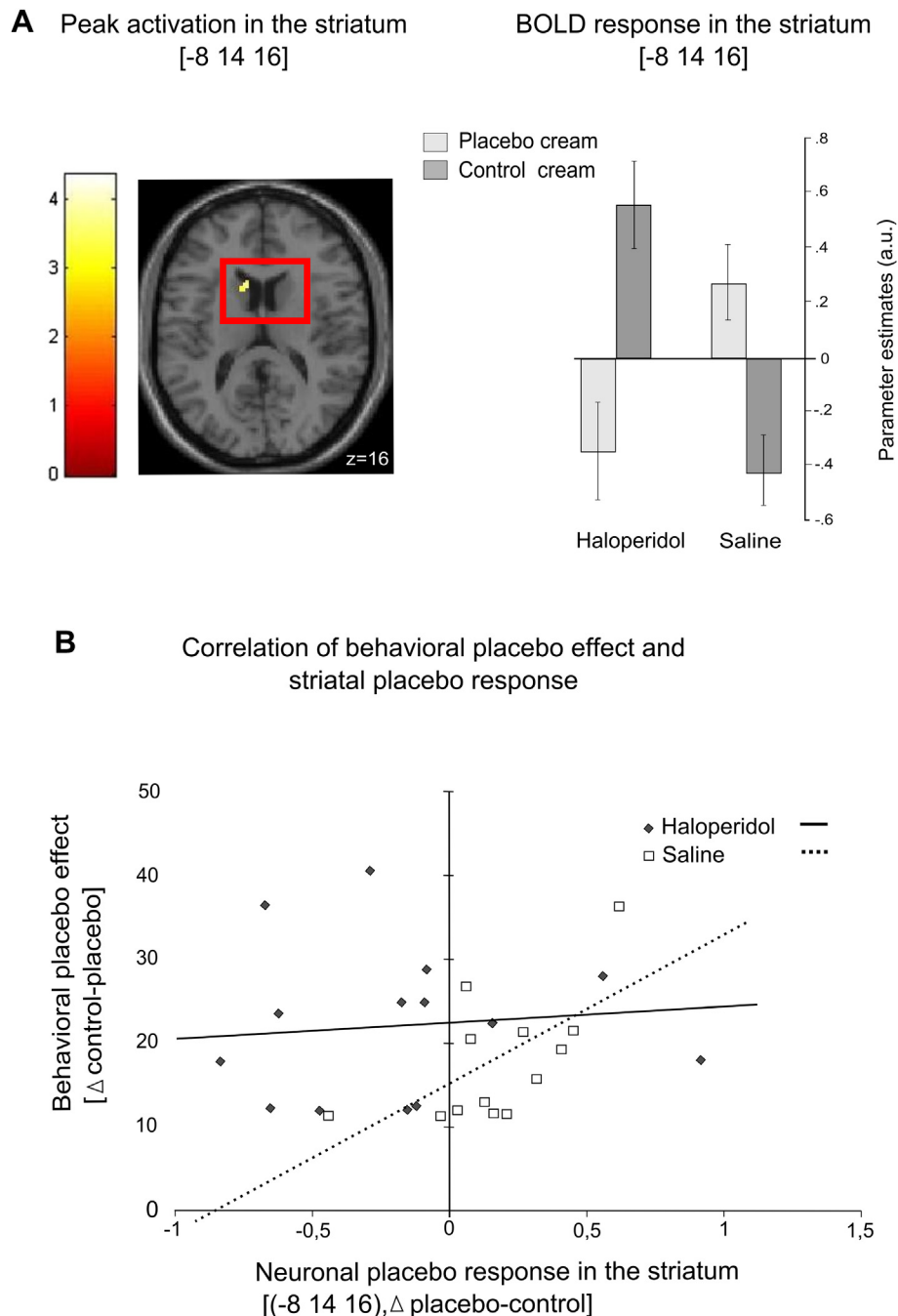


Fig. 5 – Haloperidol specifically blocks placebo-related activity in the striatum: (A) BOLD activation related to the ‘group-by-condition’ contrast [$\text{saline}_{[\text{placebo} > \text{control}]} > \text{haloperidol}_{[\text{placebo} > \text{control}]}$] and peak voxel parameter estimates reveal that haloperidol specifically blocks placebo-related activity in the striatum. The image is thresholded at $p < .001$ uncorrected and overlaid on a T1-weighted sagittal slice. The color bar indicates t-scores. Error bars indicate standard error of the mean. **(B)** Correlation between the behavioral placebo effect and placebo-related striatal activity for both groups. In the saline group, the behavioral placebo-analgesic effect is positively correlated with placebo-related activity in the striatum. This effect is abolished in the haloperidol group.

condition: $F_{(1,36)} = 56.7, p < .001$, Fig. 2). The group-by-condition interaction did not reach statistical significance ($F_{(1,36)} = .1, n.s.$), suggesting that a blockage of D2/D3 dopamine receptors did not reduce the placebo-analgesic response (as compared to the saline condition). Furthermore, haloperidol had no effect on pain thresholds ($F_{(1,35)} = .19, n.s.$) and temperatures used to induce a VAS score of 60 in the test phase ($t_{(24)} = .13, n.s.$).

3.1.4. Side effect profiles and mood scales

The haloperidol group scored significantly higher on the 7-item questionnaire regarding physical side effects (sum score, see Section 2) compared to the saline group ($t_{(36)} = 2.2, p < .01$).

On the 16-item mood questionnaire, the haloperidol group showed significantly lower scores in factor ‘alertness’ and ‘calmness’ compared to the saline group ($t_{(36)} = 1.72$; $t_{(36)} = 1.75, p < .05$). Accordingly, side effect profiles and mood scales revealed behavioral changes in the haloperidol group and in combination with verified haloperidol plasma levels indicated a successful pharmacological modulation of the dopaminergic system.

3.1.5. Reaction times

All participants were generally slower in the test phase compared to the conditioning phases (main effect of time: $F_{(1,35)} = 6.1, p < .05$). However, haloperidol had no significant effect on response times as indicated by the non-significant group difference and interaction of the reaction times (interaction: $F_{(1,35)} = .66, n.s.$; main effect of group: $F_{(1,35)} = .2, n.s.$).

3.2. fMRI results

3.2.1. Placebo-related activity

The main effects of condition revealed changes in brain activity as previously reported for placebo analgesia. The contrast [placebo > control] revealed increased activity in the DLPFC and the rACC, essential parts of the descending pain-modulatory system, in the early phase of the painful stimulus (Table 2A). The reverse contrast [control > placebo] (independent of group) revealed increased pain-related activity in the contralateral posterior insula, the contralateral somatosensory cortex and the dorsal anterior cingulate cortex (dACC) in the control compared to the placebo condition, particularly in the late phase (Table 2B).

Haloperidol did not affect any of these placebo-related activity changes in pain sensitive or pain-modulatory areas, as revealed by the interaction analysis (Table 2C and D) of the entire pain period as well as parameter estimates of each pain phase and group separately (Fig. 4). However, haloperidol significantly reduced placebo-related activity in the caudate nucleus as revealed by the interaction analysis of the entire pain period (Fig. 5A, Table 2C).

Post-hoc analyses carried out to clarify the behavioral significance of this finding revealed a significant positive correlation between the behavioral placebo effect and the neuronal placebo effect (defined as the difference in parameter estimates between placebo to control site) in the saline group ($r = .62, p < .05$), but not in the haloperidol group ($r = .13, n.s.$, Fig. 5B).

Table 3 – BOLD responses for the main effect of group.

Region	Coordinate of peak voxel (x, y, z in mm)		Voxel-level (T)	Number of voxels in cluster
	R: right	L: left	R/L	
N = 38				
(A) Saline > haloperidol				
Inferior parietal lobule	58, −56, 38		4.1	82
VTA/SN	12, −6, −10		4.1	40
Dorsal insula	42, −20, 26		3.6	18
DLPFC	44, 26, 34		3.5	15
Thalamus	10, −6, 4		3.5	9
PAG	6, −26, −20		3.2	1
Globus pallidus		−12, 4, −6	3.2	3
(B) Haloperidol > saline				
—	—	—	—	—
N = 29				
(A) Saline > haloperidol				
Inferior parietal lobule	58, −56, 38		4.4	47
Insula	38, 8, −10		3.9	11
Thalamus	20, −24, 10		3.7	27
Putamen/insula	30, 14, 10		3.7	8
Middle frontal gyrus	30, 56, 26		3.7	33
	18, 46, 44		3.4	10
VTA/SN	12, −6, −10		3.7	17
DLPFC	46, 26, 34		3.5	15
PAG	10, −24, −20		3.3	1
(B) Haloperidol > saline				
—	—	—	—	—
Changes in BOLD responses to painful thermal stimulation in the control condition for the indicated contrasts. Coordinates are denoted by x, y, z in mm (MNI-space) and strength of activation is given as t-scores [(a): df = 35; (b): df = 26]. All $p \leq .001$ uncorrected, as indicated in Section 2. DLPFC: dorsolateral prefrontal cortex, PAG: periaqueductal gray, VTA/SN: ventral tegmental area/substantia nigra.				

Table 4 – Placebo-related BOLD responses for the interaction analysis in the early and late pain phases (N = 29).

Region	Coordinate of peak voxel (x, y, z in mm)		Voxel-level (T)	Number of voxels in cluster
	R: right	L: left	R/L	
(A) Early pain: interaction contrast [Saline] _[placebo>control] > [haloperidol] _[placebo>control]	—	—	—	—
(B) Late pain: interaction contrast [Saline] _[control>placebo] > [haloperidol] _[control>placebo]	—	—	—	—
(C) Late pain: interaction contrast [Saline] _[placebo>control] > [haloperidol] _[placebo>control]	—	—	—	—
Caudate body	—	–10, 14, 14	3.1	6*
(D) Entire pain period: interaction contrast [Saline] _[control>placebo] > [haloperidol] _[control>placebo]	—	—	—	—

Changes in BOLD responses to painful thermal stimulation for the indicated contrasts. Coordinates are denoted by x, y, z in mm (MNI-space) and strength of activation is expressed in t-scores [df = 54]. *p = .051 using SVC as indicated in Section 2.

3.2.2. Main effect of group

To further explore the modulatory effect of dopaminergic neurotransmission on pain processing at the neural level irrespective of the influence of a placebo, we also tested for changes in activity in the haloperidol group relative to the saline group, including all participants (N = 38). This analysis showed significantly reduced activity under haloperidol in brain regions with high dopamine receptor density belonging to the dorso-lateral frontostriatal circuit such as the contralateral DLPFC, the contralateral dorsolateral caudate nucleus, parts of the ipsilateral globus pallidus, the contralateral ventral tegmental area/substantia nigra (VTA/SN) and the contralateral thalamus [saline > haloperidol; Table 3A, Fig. 6]. The reverse contrast [haloperidol > saline] did not reveal any significant activation.

4. Discussion

In this study we explored the role of dopamine in placebo analgesia. We used pharmacological fMRI in combination

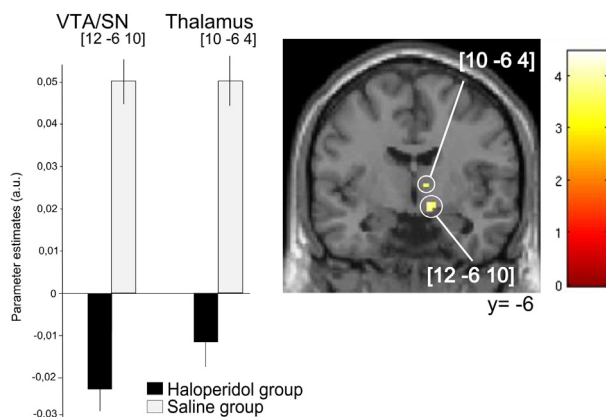


Fig. 6 – Main effect of group: BOLD activation related to the main effect of group [saline > haloperidol] and peak voxel parameter estimates for N = 38. The contrast [haloperidol > saline] did not reveal any significant activation. For visualization purposes the image is thresholded at $p < .001$ uncorrected. Color bar indicates t-scores. Error bars indicate standard error of the mean.

with a well-established placebo analgesia paradigm in healthy volunteers and blocked dopaminergic neurotransmission using 2 mg of the D2/D3 receptor antagonist haloperidol p.o. in a double-blind design.

Our placebo manipulation induced a significant placebo-analgesic effect on the behavioral level, as indicated by the difference in pain intensity ratings on the placebo-treated compared to the control site. At the neural level, pain relief during placebo analgesia was associated with decreased activity in pain processing regions such as the posterior insula and somatosensory cortices and an increased engagement of the descending pain-modulatory system, including the PFC and rACC. These neural findings are in accordance with previous neuroimaging studies of placebo analgesia (Bingel et al., 2006; Eippert et al., 2009; Wager et al., 2004).

The 2 mg haloperidol p.o. dosage induced a successful modulation of dopaminergic neurotransmission evident on both the behavioral and the neural level. Mood changes and physical side effects were more pronounced in participants who received haloperidol than in controls (Fig. 3). At the neural level, we found significantly decreased responses to pain in brain regions characterized by high dopamine receptor density such as the VTA/SN (Missale, Nash, Robinson, Jaber, & Caron, 1998) and frontal regions, including the medial prefrontal and orbitofrontal cortices (O'Doherty, 2004; Pleger et al., 2009; Zald et al., 2004).

Placebo-related activity in the dorsal striatum was positively correlated with the behavioral placebo-analgesic response in the saline group, but not in the haloperidol group. This is in line with previous findings showing a link between striatal activation (Scott et al., 2007, 2008) or gray matter density (Schweinhart et al., 2009) and the individual placebo response. The reduction of placebo-related striatal activity under haloperidol did, however, not alter the placebo-analgesic response. Haloperidol neither affected behavioral indicators of placebo analgesia (i.e., pain intensity ratings), nor placebo-related changes in somatosensory or top-down pain-modulatory regions. Together, this suggests that although the signal level in the dorsal striatum scales with the individual placebo response, dopaminergic neurotransmission is unlikely to be causally involved in the pain alleviating effect of placebo analgesia. The finding that the blockage of

dopaminergic striatal activity does not hamper placebo analgesia, raises the question why we observed a close correlation between dopamine-dependent striatal activity and the individual placebo response.

The dopaminergic system is involved in the regulation of various cognitive, behavioral and sensorimotor functions and particularly in reward mechanisms (Schultz, 1998; Wise, 1980). Of these functions, placebo analgesia has most commonly been related to reward processing, as the expectation and experience of pain relief is inherently rewarding (de la Fuente-Fernández et al., 2001; Scott et al., 2007). The dopamine-dependent striatal activity observed in our study might therefore be related to reward processes that are innately linked with the expected and perceived pain relief during placebo analgesia, and would explain the scaling of striatal response and analgesic benefit. The dosage of 2 mg haloperidol p.o. used in our study has previously been shown to induce a successful modification of reward-related responses in behavioral and neural measures in healthy volunteers (Pessiglione et al., 2006; Pleger et al., 2009; Tremblay, Desmond, Poulos, & Zack, 2011). However, as we did not assess reward-related processes explicitly, we have only indirect evidence for this interpretation.

Using [^{11}C]-labeled raclopride Scott and colleagues showed that regional dopamine activity in the ventral basal ganglia, including the striatum is associated with the anticipated and perceived effectiveness of a placebo treatment in healthy volunteers (Scott et al., 2007, 2008). To further explore the role of striatal dopaminergic activity observed during the anticipation of placebo analgesia they combined a PET investigation of placebo analgesia and fMRI measurements of monetary reward paradigm in the same group of participants. Scott et al. found a correlation between dopaminergic responses in the striatum in the anticipation phase of placebo analgesia and BOLD activity of the NAc during the expectation of monetary reward in the fMRI experiment (Scott et al., 2007). Positive correlations were also found between the BOLD activation during reward anticipation and the placebo-analgesic effect during pain trials, accounting for 28% of the variance in formation of placebo analgesia. This finding suggested that variations in dopaminergic reward processing might be related to one's individual placebo-analgesic responsiveness. Our finding that blocking placebo-related striatal activity did not impact placebo analgesia renders this hypothesis unlikely.

It should, however, be noted that the placebo-related activity that was blocked by haloperidol in our study was located in a slightly more dorsal part of the striatum compared to the placebo-related changes in dopaminergic neurotransmission found in previous molecular imaging studies (Scott et al., 2007, 2008) and also compared to reward-related activity in other studies [i.e., monetary reward (Volkow et al., 2002; Zald et al., 2004)]. The two dopamine-ligand PET studies (Scott et al., 2007, 2008) found placebo-related changes in the NAc. This discrepancy in findings might be related to the fact that fMRI is less sensitive to tonic components than PET. Intriguingly, dopamine release in the dorsal striatum (i.e., caudate and putamen) has previously been found to correlate with the clinical benefit (motor improvement) in patients with Parkinson's disease following a subcutaneous injection of saline introduced as apomorphine (de la Fuente-Fernández et al.,

2001). This observation may support the notion of the dorsal striatum being linked to the generation of placebo responses independent of the specific system (i.e., pain, motor) involved.

Our interpretation that dopaminergic activity is not directly involved in the generation of analgesia in a placebo paradigm is further supported by the fact that we did not observe a main effect of haloperidol on general pain sensitivity, as indexed by pain thresholds or temperatures yielding a pain intensity of 60 on the VAS. Although a role of dopaminergic neurotransmission in pain sensitivity and modulation has been debated, available animal and human studies have revealed inconclusive results. In animal studies, intraventricular or striatal microinjection of haloperidol has been shown to increase acute pain, whereas injections of the dopamine agonist apomorphine reduced noxious response (Lin, Wu, Chandra, & Tsay, 1981; Magnusson & Fisher, 2000). These experimental findings are complemented by clinical data of patients suffering from diseases involving pathologies of the nigrostriatal dopaminergic system such as Parkinson's disease or schizophrenia. PET studies in healthy volunteers have revealed a low dopamine D2 receptor availability in the putamen that is associated with increased cold pain thresholds (Hagelberg et al., 2002). These findings support the notion that an increase in dopaminergic neurotransmission promotes antinociceptive processes whereas a lack of dopaminergic transmission might have pronociceptive effects, resulting in a hypersensitivity to pain or a higher prevalence of clinical pain conditions. However, the changes in pain sensitivity in Parkinson's patients could also be explained by the neurodegeneration occurring in serotonergic and noradrenergic systems which is pathognomonic for Parkinson's disease (Moore, Whone, & Brooks, 2008; Taylor et al., 2009). This would explain the inconsistent results regarding changes in pain sensitivity when levels of dopaminergic transmission are pharmacologically modulated in this patient group (Djaldetti et al., 2004; Gerdelat-Mas et al., 2007; Gierthmühlen et al., 2009). A recent study modulating dopamine levels using the dopamine agonist sulpirid and aminoacid depletion in young healthy volunteers also found no effects on different models of experimental pain (Becker et al., 2013).

These and our findings suggest that the relationship between dopaminergic neurotransmission, pain sensitivity and pain modulation is complex and further research is needed to elucidate its determinants. A key point in this search is the fact that dopamine is involved in a number of processes that are inherent to pain perception but not specific. For instance, in a placebo analgesia paradigm the participant constantly monitors the perceived intensity of a painful stimulus and compares this perception to the expected pain relief before an individual judgment about perceived pain is made on a trial-by-trial basis. Both processes have consistently been associated with dopaminergic neurotransmission in the striatum, including dorsal aspects (Cooper, Dunne, Furey, & O'Doherty, 2012; Haruno & Kawato, 2006; O'Doherty et al., 2004).

Although the dosage of 2 mg haloperidol p.o. used in our study has previously been shown to modulate reward-related responses in behavioral and neural measures in healthy volunteers (Pessiglione et al., 2006; Pleger et al., 2009; Tremblay et al., 2011), it is conceivable that reward-related processing was not modulated sufficiently enough in our sample to also affect placebo analgesia. Future studies are therefore needed

to investigate dose-dependent effects on both behavioral and neural responses underlying reward-related processing and pain modulation in the same sample.

5. Conclusion

Taken together, we show that 2 mg of the D2/D3-antagonist haloperidol exclusively blocks placebo-related activity in the striatum, but does not affect behavioral and neural measures of placebo analgesia. Our study therefore suggests that dopaminergic neurotransmission might not be causally involved in analgesia, but rather reflects other dopaminergic processes related to placebo analgesia. A more detailed understanding of their role could provide important insights into the way the brain helps or hinders the unfolding of a placebo response.

Conflict of interest

The authors declare no conflicts of interest.

Acknowledgement

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